Shahir Ahmed

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EDUCATION

Michigan State University, East Lansing, MI **GPA: 4.00**
Bachelor of Science, ***Electrical Engineering****, (Minor in Mathematics)* Expected May 2029
Affiliations: IEEE MSU Student Branch **(Technical Director)**

EXPERIENCE

MSU Solar Car Racing East Lansing, MI
Harnessing and Integration Lead *September 2025 – Present*

- Developed and deployed STM32 firmware in C++ for an on-vehicle controller, executing real-time CAN and LIN communication for 5 vehicle signals and faults, including turn signals, hazards, braking, and battery protection faults
- Engineered safety-critical vehicle control electronics in 4-layer PCB using Altium Designer, validating 25+ MCUs, sensors, MOSFETs, and regulators across automotive operating ranges (-40°C to +105°C, 9-36V)
- Redesigned and rebuilt harness architecture across 20+ interconnects for BMS, motor drive, and control systems under expedited timeline, eliminating 100% of recurring faults and achieving automotive HV isolation compliance
- Guided vehicle through 11 scrutineering inspections and managed trackside pit strategy at FSGP 2026, sustaining 120+ miles over 5 hours of racing to secure top 20 finish, achieving Michigan State's best result in 12 years

RF-Photonics Lab, Michigan State University East Lansing, MI
Undergraduate Researcher *June 2026 – Present*

- Programmed 50+ GNU Radio flowgraphs and native UHD Python transmit/receive software implementing FIR filtering, pulse shaping, symbol-timing recovery, and QPSK/16-QAM/256-QAM modulation on USRP B210
- Cut square-wave rise/fall time from 135 ns to 45 ns on signal generator by tuning sample rate, frequency, and gain, characterizing link quality with BER and EVM and resolving underflow and cable-length induced phase offsets
- Transmitted OFDM data and an LFM chirp concurrently in same frequency band at 3±0.005 GHz, demonstrating simultaneous communication and radar sensing on shared spectrum, transferring 20MB error-free data in single run
- Calibrating PID-based laser frequency-lock feedback loop, writing Python control software over RS-232 to drive laser analog frequency-control registers for source stabilization within 1.2% of reference frequency

Microtechnology Lab, Michigan State University East Lansing, MI
Undergraduate Researcher *September 2025 – Present*

- Conducted 1,000+ electrochemical experiments detecting heavy metal ions down to 10 ppb using cyclic voltammetry, differential pulse voltammetry, and fast-scan CV with carbon fiber and boron-doped diamond working electrodes
- Debugged a 4-channel potentiostat PCB, identifying and resolving 6 layout and assembly faults to increase functional channel yield by 100% and fabricated microfluidic platform to further increase experimental efficiency by 25%
- Analyzed 100+ experimental datasets in Origin, MATLAB and internal electrochemical analyzer, producing graphical outputs to benchmark electrode performance across 10+ samples and validate system sensitivity thresholds
- Modeled 3D COMSOL simulation of 5mm microfluidic channel, solving laminar flow and species concentration to map post-mixing concentration maxima, validated against measured sensor response within 2% error margin

PROJECTS

- **CAN Spy:** Designed and assembled a multi-protocol bus analyzer on STM32C552, monitoring and transmitting on CAN-FD, LIN, SPI, UART, and I3C, streaming decoded traffic to a host-side GUI used as team's debug tool
- **Electrochemical Analysis Toolkit:** Built automated Python voltammetry analysis toolkit with PySide6 GUI, reducing curve processing time from 10 mins to under 2 mins per run with an overall 30% increase in post-processing output
- **Scanning Electron Microscopy Research:** Characterized Lightning connector microstructure and composition via SEM (1500x) and EDS; identified 4-element material disparities to win 2nd place at 2026 MSU Research Forum
- **Wi-Fi Presence Radar**: Prototyped presence detector on ESP32 using Wi-Fi Channel State Information, detecting human motion with no dedicated sensor by processing per-subcarrier channel amplitudes at 20 frames/s

SKILLS

- **Hardware Design:** Altium Designer, KiCad, LTspice, AutoCAD, SolidWorks, Siemens NX, Schematic Capture
- **Programming & Analysis:** C, C++, Assembly, Python, MATLAB, Origin, COMSOL Multiphysics, Git, Ansys
- **Digital Design:** Verilog, RTL design, VHDL, SystemVerilog, FPGA Synthesis, Vivado, Quartus
- **Lab Equipment:** Oscilloscope, Multimeter, Signal generator, Spectrum analyzer, SDR, External Cavity Laser
- **Signal Processing & SDR:** GNU Radio, QPSK, QAM, OFDM, FMCW, LFM, BER and EVM characterization